obtained by purchasing a special imaging table with a thin Perspex base which generated minimum attenuation.

Now that tomographic imaging is almost ubiquitous, modern gamma cameras usually incorporate a built-in imaging couch of low attenuation material so that images can be taken from any direction around the patient. Sometimes the couch is an integral part of the gamma camera so that it must be used for all studies, but sometimes the couch can be completely removed so that the patient can still be imaged on a chair or a hospital bed if required.

2.4 The detector head

The detector head contains everything that is necessary to detect gamma rays emerging from the patient and produce electrical signals which represent the location of each detected event and the energy deposited in the crystal. The basic construction of the detector head of an Anger type gamma camera is shown in Figure 2.5. Most of the weight of the detector comes from the lead shielding which is placed around the back and sides to prevent gamma rays from entering from this direction. The shielding may need to be several centimetres thick to provide adequate attenuation of the highest gamma ray energies likely to be encountered. In normal use, the front of the detector is covered by a collimator.

2.4.1 The collimator

The collimator consists of a thick lead plate with many narrow holes in it. This blocks all gamma rays apart from those travelling along the direction of the holes, so that only a small fraction of the gamma rays incident at the front of the collimator passes through. Unfortunately, this poor efficiency is necessary to form an image on

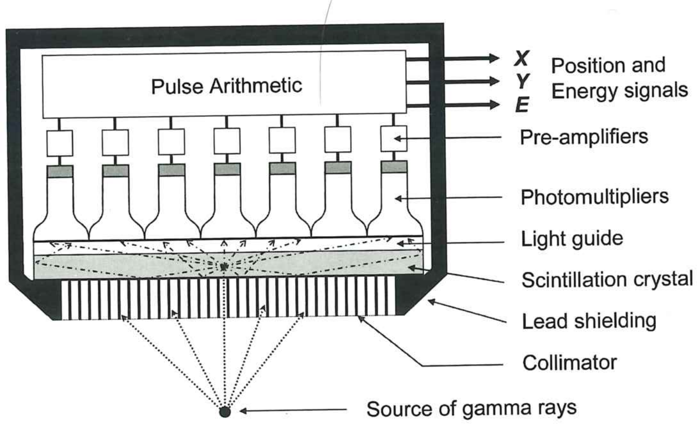

Figure 2.5 Components in the detector head of an Anger gamma camera

the scintillation crystal of the radioactivity distribution within the source. The collimator therefore performs the same function as the lens in an ordinary photographic camera. Different types of collimators are used for different applications and a detailed discussion of collimator design can be found in Chapter 4.

2.4.2 The scintillation crystal

If the collimator is like the lens of a photographic camera, then the scintillation crystal is like the film and the associated electronics serve a purpose equivalent to developing the film to make the image visible. This happens in several stages. First the gamma radiation is converted to visible light by the scintillation crystal. However, this is much too weak to be seen by the naked eye and so the light is converted to electrical signals and amplified by an array of photomultiplier tubes. These are optically coupled to the rear face of the crystal *via* a transparent light guide. Figure 2.6 shows the scintillation crystal taken from a gamma camera with some photomultiplier tubes in position. The crystal and light guide are mounted in a circular metal ring and they can be seen where the photomultiplier tubes have been removed. Signals from the photomultiplier tubes are further amplified and passed through pulse arithmetic circuits, where they are converted to signals giving the position (X and Y) of each gamma ray detected and the energy deposited in the crystal (E).

The scintillation crystal in Anger's first gamma camera was only 100 mm in diameter and 6 mm thick, but by the 1980s, crystals as large as 500 mm in diameter were in use. A modern gamma camera may have a rectangular crystal measuring up to 500 mm × 600 mm and 9 mm thick. This is extremely costly to produce and comprises the single most expensive part of the gamma camera. However, the principle on which it operates is identical to that in any small scintillation monitor.

Figure 2.6 A scintillation crystal with its light guide and some photomultiplier tubes

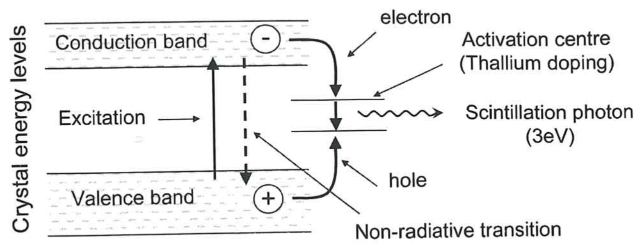

Figure 2.7 The scintillation process

Figure 2.7 illustrates the scintillation process that occurs in crystals such as thallium activated sodium iodide (NaI(Tl)). When a gamma ray interacts in the crystal, it may give up all of its energy (by photoelectric absorption) or only part of it (by Compton scattering). In either case, a secondary electron is produced which travels a short distance through the crystal gradually slowing down and causing ionisation of several thousand nearby atoms. Each ionised atom loses an electron, which is excited from the valence band into the conduction band of the crystal energy levels, leaving a hole in the valence band. Many of these electrons fall back to the valence band by non-radiative transitions which do not result in the emission of any light and just dissipate the energy as heat. However, because bound electron-hole pairs are electrically neutral, they can easily migrate through the crystal to activation centres where they fall back by radiative transfers, emitting scintillation light as they do so. It is desirable to have immediate light emission (fluorescence) rather than delayed emission (phosphorescence) and this is ensured by doping the crystal with small amounts of an activator such as thallium to create the activation centres. Fluorescent emission can occur within less than 1 μs of the gamma ray absorption, so the light pulses are very brief, which means that the detector is capable of dealing with quite high count rates. Moreover, the number of light photons emitted will be proportional to the energy deposited by the gamma ray and so the crystal makes a good energy discriminating detector.

An ideal scintillation material should have the following properties:

- high efficiency for stopping gamma rays
- high probability of photoelectric absorption rather than Compton scattering
- high conversion efficiency of absorbed energy to light
- transparent to the scintillation light which it emits
- wavelength of light emitted should match response of photomultiplier
- short scintillation time
- mechanically robust.

The scintillation crystal traditionally used in gamma cameras is made from sodium iodide with trace quantities of thallium added, and so is known as NaI(Tl). This is close to being an ideal material for the purpose. Because of its high density, it is good at stopping gamma rays and its relatively high atomic number results in a high probability of photoelectric absorption. However, the number of gamma rays that are totally absorbed within a given thickness of crystal falls with increasing gamma ray energy, because some will penetrate the crystal completely. A 9 mm thick crystal is enough to absorb 84% of 140 keV gamma rays (from ^{99m}Tc) but only 13% of 511 keV gamma rays (from positron annihilation). Increasing the crystal thickness to 16 mm will improve the detection efficiency to 95% for 140 keV and 21% for 511 keV, but only at the expense of worsening the resolution of the final image. For cameras that are mainly used for imaging low energy gamma emitters, such as ^{99m}Tc, a 9 mm thick crystal is normally used. If high energy gamma rays are also to be imaged, then a 13 mm crystal may be a suitable compromise. If 511 keV imaging is required, then 16 mm or even 19 mm thick crystals may be needed, in which case a worse resolution at low energies will have to be accepted.

The addition of traces of thallium to the sodium iodide provides activation centres in the crystal into which excited electrons are trapped (see Figure 2.7). Here electrons can recombine with the holes left in the valence band. This has three advantages. First it improves the light conversion efficiency by encouraging radiative transfers (emitting light) rather than non-radiative (thermal) transfers. Second, by creating different energy levels to those present in the body of the crystal, the thallium impurity ensures that light emission is not at a wavelength which would be absorbed by the crystal. Third, it matches the emitted wavelength nicely to the optimum response of available photomultiplier tubes. The scintillation light emitted from NaI(Tl) has a wavelength in the range 350–500 nm with a peak at 410 nm, which is in the blue part of the visible spectrum. Typically, the process is only about 10% efficient, so, on average, one light photon, with a wavelength of 410 nm (energy 3 eV), is emitted for every 30 eV of energy absorbed by the crystal. This means that when one 140 keV gamma ray (from ^{99m}Tc) is totally absorbed, about 4700 visible light photons are produced. The light pulse has a duration of about 0.25 μs, which is relatively long compared with some other scintillators, but adequate for most gamma camera applications.

The main drawback of sodium iodide is that it is fragile and the crystal cracks easily. It must also be protected from sudden temperature changes otherwise the stresses caused by differential expansion can crack the crystal. Many manufacturers specify that the temperature should not change any faster than 3 °C h^{-1}. Gamma cameras therefore are normally operated in air conditioned rooms to provide a constant temperature environment. In normal use, the collimator provides some mechanical and thermal protection to the crystal, but if the collimator is removed, the crystal becomes extremely vulnerable to damage.

Sodium iodide is also hygroscopic and it must be hermetically sealed to prevent contact with room air, otherwise it absorbs moisture from the air and the iodine dissolves and turns the crystal yellow so that it becomes opaque. A hermetic seal is achieved by encapsulating the crystal in a thin aluminium can on the front and sides, sealed to a transparent light guide at the rear which acts as an exit window. The inner surfaces of the aluminium can are covered with a diffuse reflector (such as magnesium oxide or aluminium oxide powder) so that as much as possible of the scintillation light is channelled out through the light guide. This reflective coating makes the crystal and light guide appear white in Figure 2.6, even though they are actually completely transparent.

2.4.3 The light guide

The purpose of the light guide is to form an interface between the crystal and the photomultipliers. The photomultipliers cannot be placed directly against the crystal because of the need to encase it in a hermetic encapsulation. In some early gamma cameras, the light guide was shaped to optically channel light towards the centres of the photomultipliers, but in modern cameras, it is just a parallel sided window. Nevertheless, the thickness and optical properties of the light guide can have an effect on image quality. The light guide is made of glass or quartz with a refractive index matched to the scintillator in order to minimise internal reflection and ensure that as much light as possible is passed on to the photomultipliers.

2.4.4 The photomultipliers

A photomultiplier tube (PMT) consists of an evacuated glass tube with a photocathode at the front that converts light into electrons and a series of electrodes which amplify the number of electrons. The PMTs used in a gamma camera are typically between 50 mm and 75 mm in diameter and close packed in a hexagonal pattern to ensure that minimal gaps are left between the tubes. Anger's original camera only had seven circular PMTs consisting of a central tube and a ring of six around it (see Figure 1.5). Subsequent generations of cameras with larger circular crystals added extra rings of PMTs, making 19, 37 or 61 tubes in total. Some manufacturers reduce the gaps between tubes by using hexagonal shaped PMTs. A modern rectangular detector may have more than 100 small PMTs, or fewer large PMTs. Figure 2.8 shows the layout of tubes in a typical rectangular detector. This camera has a 9.5 mm thick crystal measuring 59 cm × 45 cm, although the usable field of view is smaller than this by 3 cm at each side. It uses 59 photomultiplier tubes; 53 tubes are 75 mm across with three 50 mm tubes at each end to make an approximately rectangular shape. Each PMT is surrounded by a cylinder made of mu metal which screens it from the effects of the earth's magnetic field. The PMTs pass through holes in the white supporting plate to view the crystal which is not visible in this photo. The base of each PMT is connected to a pre-amplifier circuit with a ribbon cable taking the signals on to the camera electronics.

Figure 2.8 View of the photomultiplier tubes in a large rectangular gamma camera after the detector cover and shielding have been removed

The PMTs are stuck to the light guide with optical coupling grease which eliminates air gaps and ensures good light transmission from the crystal to the PMT. Any air gap would give a large change in refractive index and therefore increase the amount of total internal reflection. Ideally, the grease will have a refractive index midway between that of the light guide and the PMT glass. However, after several years, this grease may begin to dry out and the detector may need regreasing to maintain optimum performance. The PMTs must be kept in the dark whenever the high voltage is applied, otherwise excessive current will be drawn and the tube damaged.

The PMTs usually have bi-alkali photocathodes (e.g. caesium-antimony) deposited as a thin (20 nm) semi-transparent layer on the inner surface of the entrance window. This photocathode material is sensitive to blue light and responds by emitting photoelectrons which are focused by a wire grid and then accelerated onto a chain of dynodes held at increasingly positive voltages. The dynodes are made of a material that is a good secondary emitter (either caesium-antimony or copper-beryllium) and so more electrons are emitted at each stage, thus creating amplification of the signal. There are many different designs of PMT with different shapes of dynodes. Venetian blind type dynodes have good electron collection efficiency but a slower collection time and hence do not produce such a fast output pulse. A 'tea cup' shaped first dynode is often used because it is good at focusing photoelectrons emitted from a large area photocathode. Ideally, the PMT should have a uniform response to light falling anywhere across the face of the tube, but this is very difficult to achieve and usually the response tends to fall off at the edges due to worse collection of photo-electrons.

Figure 2.9a is a photograph of a typical photomultiplier tube and Figure 2.9b shows a schematic diagram illustrating how it works. In a real tube, the dynodes will have a more complex arrangement with different dynode shapes and possibly including separate accelerating and focusing electrodes. When a light photon enters the front